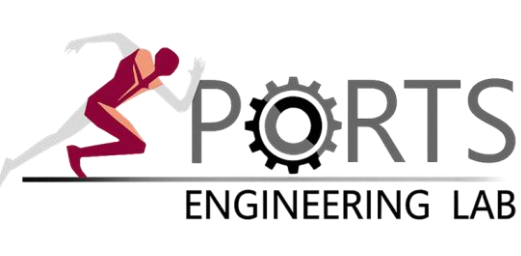


Referent-Configuration Imitation Learning (RCIL) for Predictive Musculoskeletal Simulation

Faster and more stable gait imitation by controlling referent configurations instead of muscle activations

Chihyeon Lee¹, Joeeun Ahn^{1,2,3*}, Beom-Chan Lee^{2,4*}



¹Department of Physical Education, Seoul National University, South Korea

²Institute of Sport Science, Seoul National University, South Korea

³Robot Institution, Seoul National University, South Korea

⁴Department of Health and Human Performance, University of Houston, USA

*Corresponding authors

email: chi0412@snu.ac.kr, ahnjoeeun@snu.ac.kr and blee24@central.uh.edu

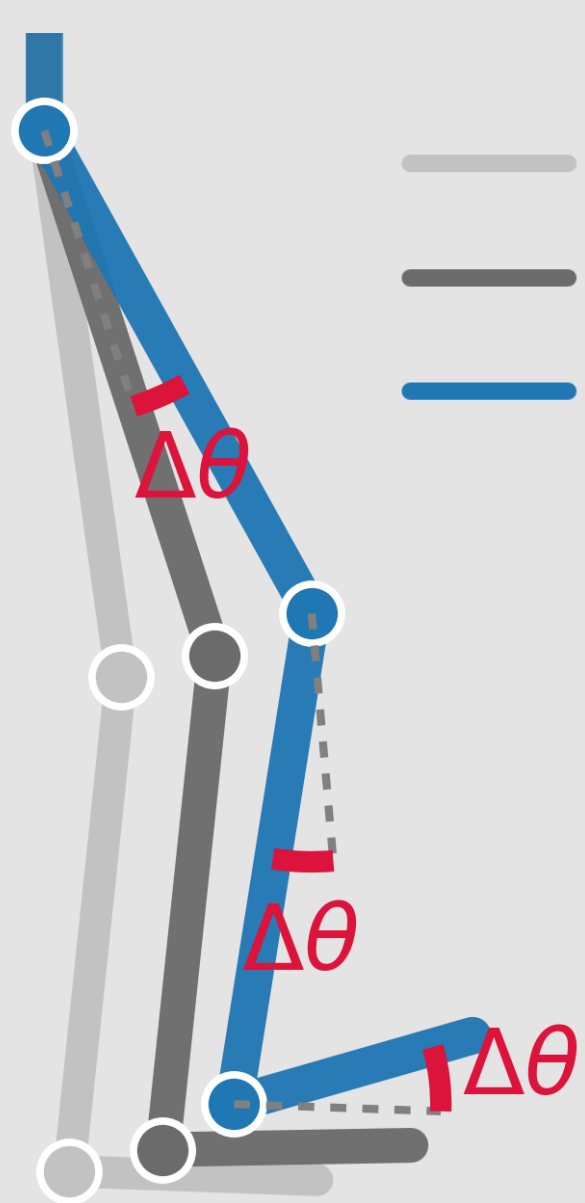
PROBLEM

- Predictive musculoskeletal simulation (MSK) is powerful, but muscle redundancy makes control hard [1].
- Imitation learning reproduces motion [2], yet controlling in muscle-activation space is slow, unstable, and prone to collapses [3].

MOTIVATION

- The CNS controls movement by shifting referent configurations (λ -thresholds), rather than muscle activations, a low-dimensional and self-stabilizing control variable [4, 5].
- Our idea: let the policy learn in referent configuration space — a small residual $\Delta\theta$ added to the reference pose.**
- The 0.1 s anticipatory lead matches the ~100 ms neuromechanical delay that the CNS compensates for [6].

METHODS & RESULTS



$$1) R = \theta_{ref}(t + 0.1s) + \Delta\theta$$

➤ R is the target referent configuration.

$$2) \lambda_R = \text{muscle fiber length}(R)$$

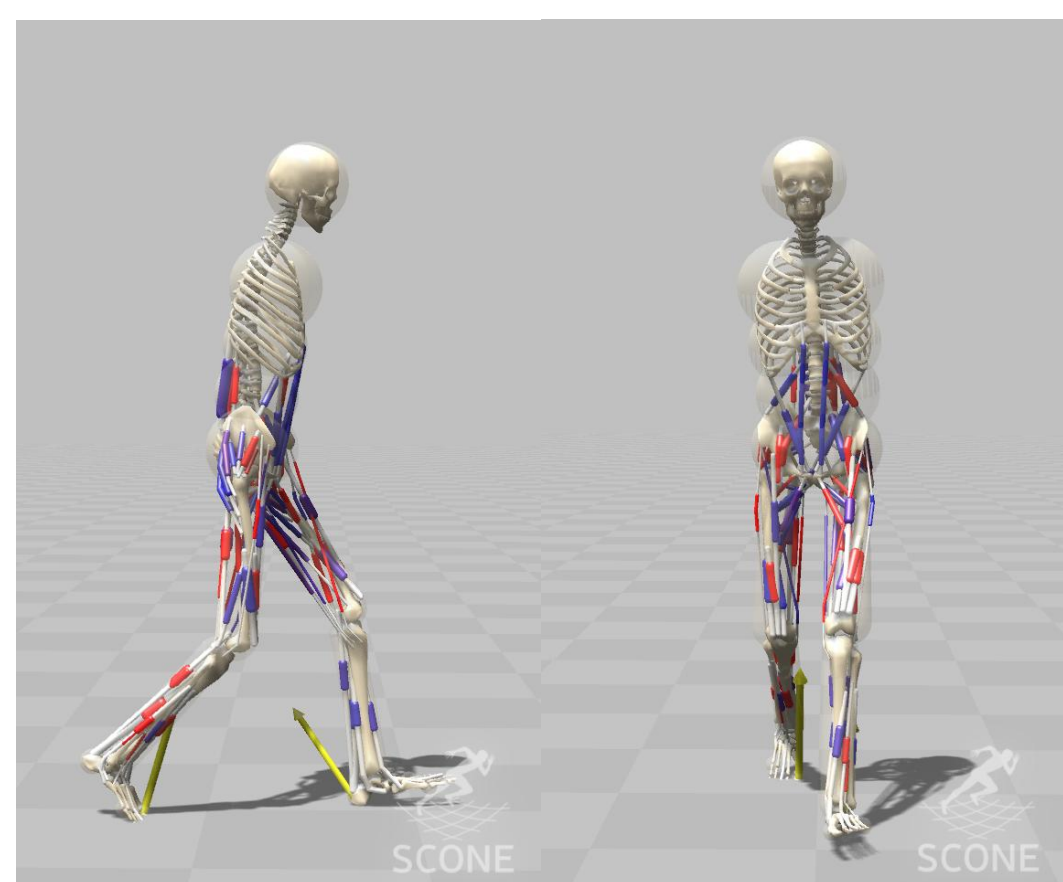
➤ λ_R is the muscle fiber length at R .

$$3) A(t) = [g(L(t) - (\lambda_R - \delta) + \mu V(t))]_0^1$$

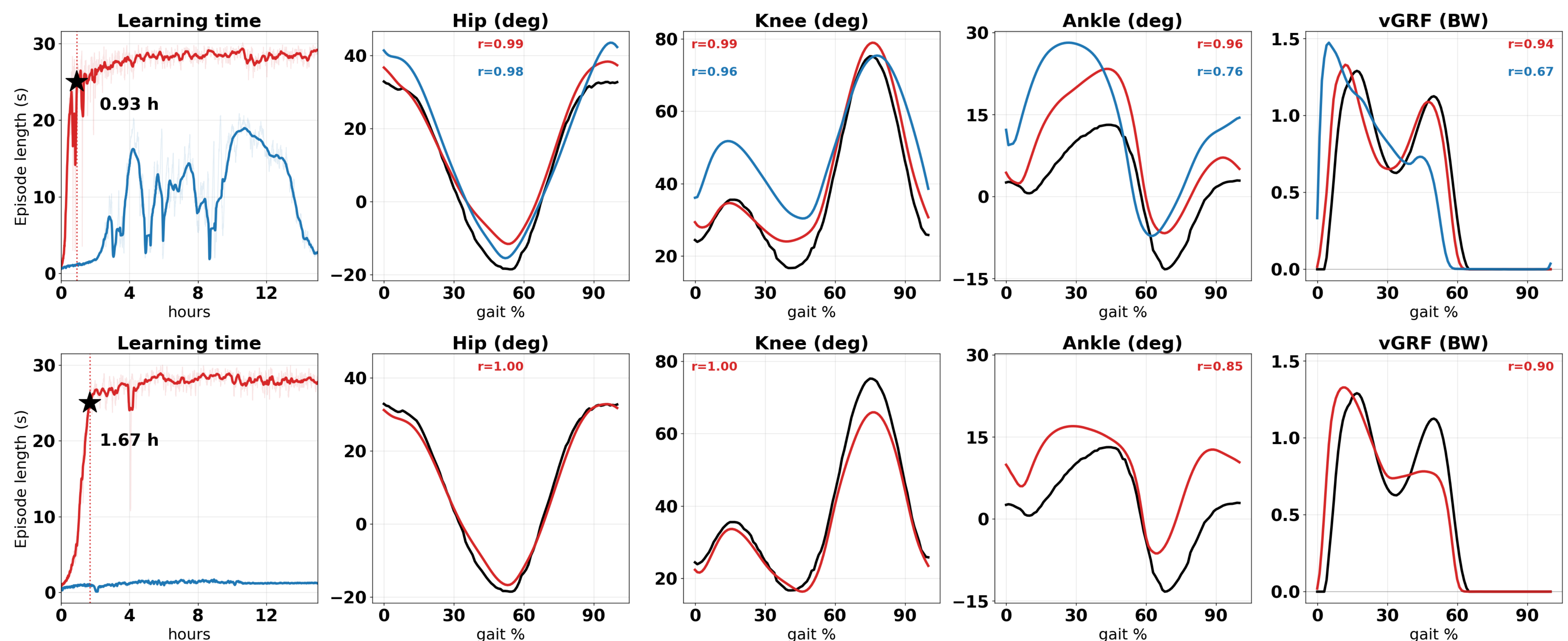
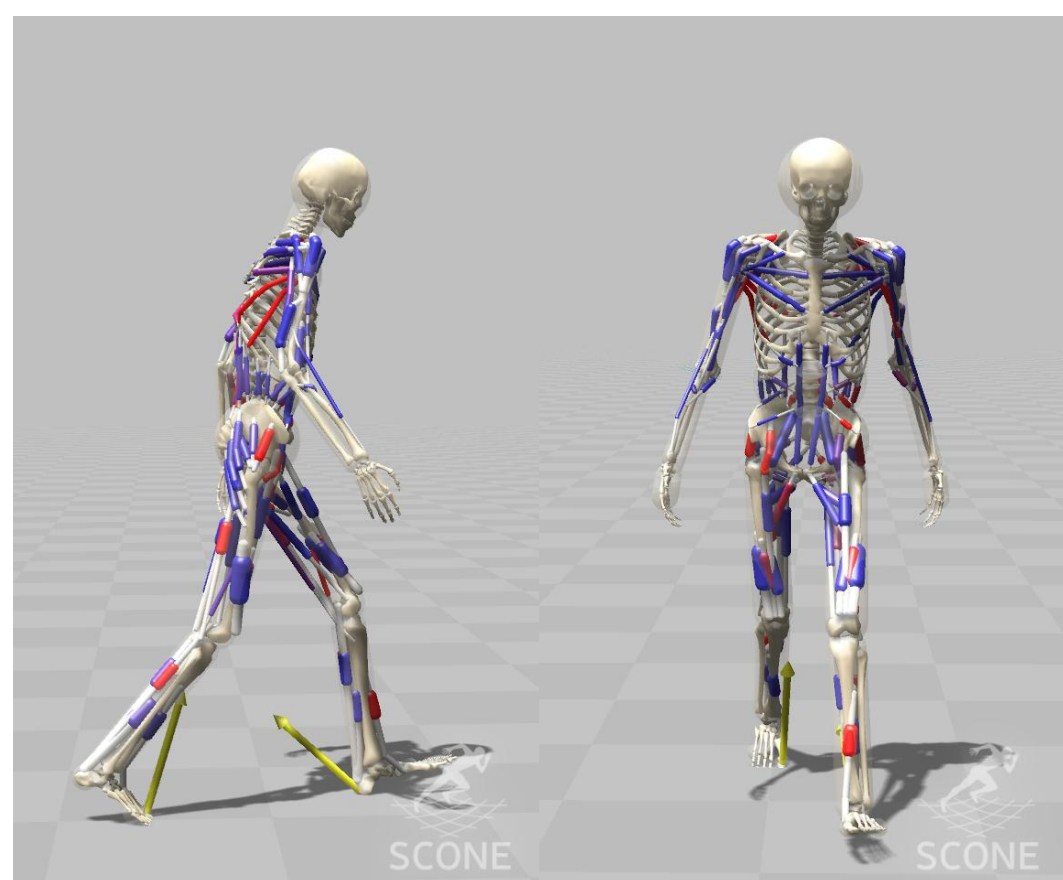
➤ $A(t)$ is the muscle activation [4, 5].

- $L(t)$: Muscle fiber length
- $[g]_0^{20}$: Length-based reflex (tonic) gain
- $[\delta]_0^{0.2}$: Baseline muscle tone
- $V(t)$: Muscle fiber Velocity
- $[\mu]_{-0.1}^{+0.1}$: Velocity-based reflex (phasic) gain
- $(\lambda_R - \delta)$: Length-threshold

H2190 (SCONE)



H32130 (SCONE)



Check
this out!

- RCIL reached stable walking within 1-2 hours in both models (H2190 0.93 h, H32130 1.67 h), whereas pure IL catastrophically forgot (H2190) or never converged (H32130).**
- RCIL tracked reference joint kinematics closely across both models (hip & knee $r \geq 0.99$, ankle $r = 0.96/0.85$), while pure IL degraded distally (H2190 ankle $r = 0.76$).**
- RCIL reproduced the vertical GRF ($r = 0.94/0.90$), far above pure IL, which lost push-off (H2190 vGRF $r = 0.67$).**

DISCUSSION & CONCLUSION

- Commanding referent configurations instead of muscle activations turns muscle redundancy into a low-dimensional, dynamically consistent target, so RCIL converges quickly to stable walking while direct imitation does not.
- RCIL turns motion data into subject-specific musculoskeletal digital twins that are fast to train, accurate, and ready for predictive simulation.

ACKNOWLEDGEMENT

Brain Pool Program (RS-2024-00446461) funded by the Ministry of Science and ICT through the National Research Foundation of Korea, and Korea Health Technology R&D Project through the Korea Health Industry Development Institute (KHIDI) funded by the Ministry of Health & Welfare (No. HK23C0071).

REFERENCES

[1] Geijtenbeek et al. (2019), SCONE, [2] Peng et al. (2018) [3] Song et al. (2021), [4] Feldman (1986), [5] St-Onge (2004) [6] Afschrift et al. (2021)